

INDEXING AND ABSTRACTING

A Beginner-Friendly Guide to LIS 815

Study Guide Edition

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This study guide is an original educational compilation prepared to support students taking LIS 815. It draws on the LIS 815 and LIS 814 course outlines and on standard reference works in the field, and is intended for use as a study aid alongside prescribed texts and lecture materials. Where this guide differs from the instructions of a lecturer or the requirements of an institution, those instructions and requirements take precedence.

This expanded edition incorporates additional material on book indexing, the digital and AI-assisted transformation of indexing practice, and further scholarly definitions and citations, while preserving the structure, tone and pedagogical approach of the original edition.

Preface

This book has been written for postgraduate students who are new to the subject of indexing and abstracting. Plain language is used throughout, while retaining the technical terms required for the course.

The guide follows a fourteen-chapter structure organised into seven modules, and places emphasis on understanding before memorisation. Each chapter includes learning objectives, explanations in everyday language, worked examples, and review questions to help consolidate learning.

For systems such as PRECIS, students should follow the exact operator rules prescribed by their lecturer or course manual, since notational conventions can vary between institutions.

The structure of this guide reflects common assessment components in courses of this kind: thesaurus construction and PRECIS practice, abstract writing, comparison of pre-coordinate and post-coordinate systems, and retrieval calculations.

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MODULE 1 — THEORETICAL FOUNDATIONS AND TYPES OF INDEXES

Chapter 1: Fundamentals and Objectives of Subject Indexing

By the end of this chapter, you should be able to:

- Define subject indexing and subject analysis.
- Explain why indexing is necessary for information retrieval.
- Identify the main objectives and guiding principles of subject indexing.
- Relate indexing decisions to the information needs of users.

1.1 What Is Subject Indexing?

Subject indexing is the process of representing what a document is about by assigning terms, phrases, symbols, or other representations that a person can later use to find that document. In simple terms, subject indexing answers two connected questions: what is this document about, and which words or terms should a user search for to find it?

Think of a library, a database, or even a search engine as a very large room full of documents. Without a system that labels each document by its subject matter, a user would have to read every single item to know what it contains. Subject indexing is the labelling system that makes this unnecessary.

Several scholars have refined this basic idea. Lancaster (2003) defines subject indexing as the process of determining the subject content of a document and assigning terms that adequately represent that content for retrieval purposes. Rowley and Farrow (2000) describe it as the assignment of descriptive terms that represent the topics discussed in documents so that they can be identified during information searches. The British Standard BS ISO 5963 frames indexing as the process of examining documents, identifying their subject matter, and selecting indexing terms that accurately represent their contents, while ISO 999 similarly treats indexing as the systematic analysis and representation of the intellectual content of documents to facilitate retrieval. Taken together, these definitions point to three recurring components of subject indexing: subject analysis, selection of indexing terms, and information retrieval.

1.2 What Is Subject Analysis?

Subject analysis is the thinking stage that happens before any term is chosen. The indexer reads and examines the document to identify its main topic, the important concepts it discusses, how those concepts relate to one another, and which aspects of the document will matter most to the people who will eventually search for it. An indexer performing subject analysis does not simply copy every word that appears in the title; instead, the indexer interprets the content and decides what truly represents it.

EXAMPLE

A paper titled “Use of Artificial Intelligence in Nigerian University Libraries” is not just about the words in its title. Careful subject analysis might identify the core concepts as Artificial Intelligence, University Libraries, Nigeria, Library Services, and Information Technology. Which of these are chosen as index terms depends on the indexing language and policy in use.

1.3 Why Indexing Matters

Large collections of documents contain far more material than any user could inspect item by item. Indexing creates organised access points that connect specific documents to the information needs of

specific users, making systematic and efficient retrieval possible. Without indexing, search would depend entirely on chance or on browsing everything.

1.4 A Short History

Early systems of retrieval relied mainly on access by author and by title. As collections grew larger, it became clear that subject access was also required, since users often know what a document is about before they know who wrote it. Over time, several traditions developed to provide this subject access, including formal subject headings, classification-based indexing, chain indexing, coordinate indexing systems, and, more recently, computer-assisted and machine-supported methods.

1.5 How Users Actually Search

Users rarely search using the exact words that an author or an indexer might choose. They may use synonyms, broader or narrower terms, informal phrases, or natural, everyday language that differs from the technical vocabulary of the field. Good indexing practice, supported by controlled vocabularies, helps bridge the gap between how users express their needs and how documents are actually labelled.

1.6 The Main Objectives of Subject Indexing

The major objectives of subject indexing can be summarised as follows:

- Accurate subject representation — making sure the index terms genuinely reflect what the document is about.
- Consistent access points — using the same term for the same concept every time, so that related documents are grouped together.
- Efficient retrieval — helping users find what they need with the least possible effort.
- Reduction of ambiguity — avoiding confusion between terms that sound alike but mean different things, or between different terms that mean the same thing.
- Connection of related concepts — showing how ideas relate to one another so that users are not limited to a single narrow search.
- Support for user discovery — helping users find documents they did not already know existed.
- Improved precision and recall — helping ensure that searches return documents that are genuinely relevant, without missing too many that are.

1.7 Guiding Principles of Subject Indexing

Beyond its objectives, professional practice is guided by a set of working principles that an indexer applies to every document:

- Specificity — indexing terms should describe the subject as precisely as the vocabulary allows.
- Consistency — the same concept should always be represented using the same authorised term.
- User orientation — terms should reflect the language and search behaviour of the people who will actually be searching.
- Exhaustivity — all significant subjects discussed in the document should be represented.
- Selectivity — only meaningful and genuinely relevant concepts should be indexed, not every word that happens to appear.
- Objectivity — indexing should reflect the document's actual content rather than the personal opinions of the indexer.

These principles reappear throughout this guide, particularly in Chapter 10, where exhaustivity and specificity are treated as measurable dimensions of indexing quality.

1.8 A Note on Assigned Indexing

When an indexer selects terms from a controlled vocabulary or indexing language rather than copying words directly from the document, the result is known as assigned indexing. This is contrasted with derived indexing, covered in Chapter 9, where terms are taken directly from the document's own words. Most professional subject indexing in libraries and databases is assigned indexing, because it allows the indexer to represent a concept even when the author never used the preferred term explicitly.

Review Questions

1. What is subject indexing? Give one scholarly definition in addition to your own explanation.
2. How does subject analysis differ from indexing itself?
3. Why are title words alone sometimes not enough to represent a document's subject?
4. List and briefly explain four objectives of subject indexing.
5. Distinguish between exhaustivity and selectivity as guiding principles of indexing.
6. What is the difference between assigned indexing and derived indexing?

Chapter 2: Types of Indexes and Book Indexing

By the end of this chapter, you should be able to:

- Define an index and identify its main components.
- Distinguish an index from a table of contents.
- List and describe the major types of book indexes.
- Explain the stages involved in preparing a book index.

2.1 What Is an Index?

An index is a systematically arranged guide to the contents of a document, usually placed at the end of a book, journal, report, or other publication. It consists of important words, names, subjects, concepts, places, and other significant terms, arranged alphabetically, together with references — usually page numbers — that direct readers to where the information can be found.

An index serves as a pointer or roadmap to information. Instead of reading an entire publication from beginning to end, a reader can consult the index to quickly locate specific information. Knight (1979) defined an index simply as an alphabetical list of names, subjects, or other entries, with references indicating where they occur in a document. Lancaster (2003) went further, describing an index as a device that provides access to information by identifying the concepts discussed in a document and directing users to their locations. Mulvany (2005) emphasised the intellectual side of the work, defining indexing as the process of analysing the content of a document, selecting the significant concepts, and arranging them systematically to facilitate retrieval. Taken together, these definitions make clear that an index is more than a list of page numbers — it is a retrieval tool built on careful intellectual analysis.

2.2 Components of an Index

A typical index consists of four elements:

- Headings (main entries) — the primary terms or concepts under which information is listed.
- Subheadings (subentries) — more specific aspects of the main heading, used to organise related information.

- Locators — page numbers, section numbers, or paragraph numbers indicating where the information appears.
- Cross-references — See and See also references that direct readers to related or preferred terms.

EXAMPLE

Library — automation, 45–52; budgeting, 120–125; collections, 33–41; digitization, 180–195; management, 70–85; services, 95–110. Here “Library” is the main heading, and “automation,” “budgeting,” and the rest are subheadings, each with its own locator.

A good index uses See and See also with a clear division of labour: See directs the reader from a term that is not used to the preferred term (for example, E-books. See Electronic books), while See also directs the reader to a related but distinct topic that may also be worth consulting (for example, Libraries. See also Digital libraries).

2.3 Index versus Table of Contents

Although both help a reader navigate a publication, an index and a table of contents serve different purposes.

Table of Contents	Index
Appears at the beginning of a book	Usually appears at the end
Lists chapters and major sections	Lists subjects, names, places, concepts and keywords
Follows the order of the book	Arranged alphabetically
Shows the structure of the publication	Provides detailed access to specific information
Helps readers navigate chapters	Helps readers locate specific topics

2.4 Types of Book Indexes

Indexes are not all built to the same purpose. The major types found in books and other publications include:

- Subject index — the most common type, listing topics and concepts.
- Author index — arranged by the names of authors cited or discussed.
- Name index — covering persons, whether authors or not.
- Title index — arranged by the titles of works mentioned.
- Geographical index — arranged by place names.
- Chronological index — arranged by date or period.
- Citation index — showing which works have cited a given publication.
- Analytical index — covering ideas and concepts in depth, often with detailed subentries.
- Keyword index — built from significant words occurring in the text.
- Combined index — merging two or more of the above into a single alphabetical sequence.

2.5 Characteristics of a Good Index

An effective index is arranged alphabetically, is accurate and complete, uses clear and familiar terminology, includes important concepts rather than every word that appears, contains precise locators, provides cross-references to related topics, faithfully reflects the actual content of the document, and enables quick, easy retrieval of information.

2.6 Book Indexing as an Intellectual Process

Book indexing is the systematic and intellectual process of analysing the content of a book, identifying its significant concepts, subjects, names, places, events, and ideas, and arranging them in alphabetical order with accurate locators so that readers can identify, locate, and retrieve information accurately and efficiently. It is not a mechanical task of extracting words from a text; rather, it requires the indexer to understand the subject matter, determine which concepts matter to readers, select appropriate terms, and organise those terms consistently. Book indexing is particularly valued in academic, scholarly, professional, technical, legal, medical, and reference publications, where readers often seek specific pieces of information rather than reading the entire work sequentially.

2.7 The Process of Book Indexing

Preparing a book index generally follows a sequence of stages:

1. Reading the book — the indexer studies the content carefully.
2. Subject analysis — important concepts are identified.
3. Selection of entries — relevant indexing terms are chosen.
4. Preparing main entries — major concepts become headings.
5. Preparing subentries — related ideas are grouped under main headings.
6. Assigning locators — correct page or section numbers are inserted.
7. Creating cross-references — related concepts are linked with See and See also.
8. Alphabetical arrangement — entries are sorted alphabetically.
9. Editing — errors and inconsistencies are corrected.
10. Proofreading — the completed index is verified before publication.

2.8 Challenges of Book Indexing

Book indexing is time-consuming, requires genuine subject expertise, and demands consistency across what may be hundreds of entries. Indexers must select appropriate terminology, deal with interdisciplinary topics that do not fit neatly under one heading, and manage synonyms and homonyms. Professional indexing can also be costly, page numbering can shift after late layout changes, and publishing deadlines are often tight — pressures that have only intensified with the rapid growth of digital publications, discussed further in Chapter 14.

Review Questions

1. Define an index in your own words, and cite one scholarly definition.
2. List the four components of an index and give an example of each.
3. Give three differences between a table of contents and an index.
4. Name and briefly describe five types of book indexes.
5. List, in order, the ten stages typically involved in preparing a book index.
6. State three challenges commonly faced by book indexers.

Chapter 3: Semantics and Syntax in Assigned Indexing

By the end of this chapter, you should be able to:

- Explain the meaning of semantics and syntax as they apply to indexing.
- Identify equivalence, hierarchical, and associative relationships between terms.
- Explain syntactic control, role operators, and citation order.

3.1 Semantics: The Study of Meaning

In everyday language, semantics is simply the study of meaning. In indexing, semantic relationships describe how concepts and the terms that represent them relate to one another. Three relationships are especially important in indexing work: equivalence, hierarchy, and association.

3.2 Equivalence

Equivalence connects two or more different terms that can represent the same concept, or concepts that are close enough to be treated as the same for retrieval purposes. In a controlled vocabulary, one of these terms is usually chosen as the preferred term, while the others are marked as non-preferred and simply point the user toward the preferred form.

3.3 Hierarchy

A hierarchical relationship connects a general concept to a more specific one. A Broader Term (often abbreviated BT) represents the more general concept, while a Narrower Term (NT) represents the more specific one.

EXAMPLE

Libraries is a Broader Term for University Libraries; University Libraries is a Narrower Term for Libraries.

3.4 Association

A Related Term (RT) connects two concepts that are meaningfully associated with one another but are neither synonyms nor in a broader–narrower relationship. For instance, the concepts Libraries and Librarians are closely related in meaning, even though neither is a broader or narrower version of the other.

3.5 Syntax: The Study of Order and Combination

While semantics is about meaning, syntax is about how concepts are combined or ordered when they are expressed together. Good syntactic control prevents an index entry from becoming a meaningless list of separate words, and instead preserves the relationships that exist among the concepts being described.

3.6 Role Operators

Role operators identify the function that each concept plays within a subject statement — for example, distinguishing an action from the person or thing performing it, or from the object being acted upon. Role operators are especially important in structured indexing systems such as PRECIS. Because different courses and institutions may use slightly different notations, students should always learn and apply the exact operator table prescribed in their own course manual.

3.7 Citation Order

Citation order refers to the sequence in which concepts appear within a coordinated indexing expression. Changing that order can change the emphasis of the statement, and in some cases can change its meaning altogether, which is why citation order is treated as a deliberate design choice rather than an arbitrary one.

Review Questions

1. Define semantics and syntax as used in indexing.
2. Give an example of an equivalence relationship.
3. Explain the difference between BT, NT, and RT.
4. Why is syntax important in the design of an indexing system?

MODULE 2 — VOCABULARY CONTROL

Chapter 4: Thesaurus Construction: Design and Term Selection

By the end of this chapter, you should be able to:

- Define controlled vocabulary and thesaurus.
- Explain why vocabulary control is necessary.
- Select suitable candidate terms.
- Establish BT, NT, and RT relationships between terms.

4.1 What Is a Controlled Vocabulary?

A controlled vocabulary is an organised set of approved terms that are used consistently for indexing and retrieval. By limiting the number of acceptable ways to express a concept, a controlled vocabulary reduces unnecessary variation and improves the match between the words an indexer uses and the words a searcher later types. Well-known examples used in professional practice include the Library of Congress Subject Headings (LCSH), the Sears List of Subject Headings, Medical Subject Headings (MeSH), AGROVOC, the ERIC Thesaurus, and the UNESCO Thesaurus.

4.2 What Is a Thesaurus?

In information retrieval, a thesaurus is a controlled vocabulary in which the terms are connected through clearly defined relationships, such as BT, NT, and RT. This makes it more than an ordinary dictionary: a dictionary simply defines words, while a retrieval thesaurus is deliberately structured to support indexing and searching.

4.3 Selecting Candidate Terms

Candidate terms — the raw material from which a thesaurus is eventually built — may be gathered from document titles, abstracts, author-supplied keywords, full text, existing published vocabularies, reference works, and the everyday language used by the community of people who will actually be searching. Each candidate term is then reviewed for clarity, consistency, how current it is, and how useful it is likely to be for retrieval.

4.4 Preferred and Non-Preferred Terms

The preferred term is the specific form of a concept that has been selected for indexing. A non-preferred term is a synonym or near-synonym that directs both users and indexers toward the preferred form instead.

EXAMPLE

CARS USE AUTOMOBILES — here, Cars is the non-preferred term and Automobiles is the preferred term.

4.5 Building BT, NT, and RT Relationships

When building these relationships, direction matters. A narrower concept always points upward to its broader concept using BT, and a broader concept points downward to each of its narrower concepts using NT.

WORKED EXAMPLE

It would be incorrect to say LIBRARIES BT ACADEMIC LIBRARIES. The correct statement is ACADEMIC LIBRARIES BT LIBRARIES (Academic Libraries is narrower, so its Broader Term is Libraries), and correspondingly LIBRARIES NT ACADEMIC LIBRARIES.

4.6 A Complete Thesaurus Entry

A full thesaurus entry typically brings all of these relationships together in one place. For example:

EXAMPLE ENTRY

UNIVERSITY LIBRARIES — BT: Libraries. NT: Law Libraries, Medical Libraries. RT: Academic Librarians, Digital Libraries. UF: Higher Education Libraries.

4.7 The Steps in Thesaurus Construction

Building a thesaurus from scratch is itself a project with a recognisable sequence of stages:

1. Define the subject field the thesaurus will cover.
2. Collect candidate terms from the sources described in section 4.3.
3. Identify synonyms among the candidate terms.
4. Identify homonyms — terms that look alike but carry different meanings.
5. Select the preferred term for each concept.
6. Establish hierarchical (BT/NT) relationships.
7. Establish associative (RT) relationships.
8. Establish equivalence (USE/UF) relationships.
9. Arrange the terms systematically, alphabetically and/or by classified display.
10. Test the thesaurus against real indexing and searching tasks.
11. Revise and update it regularly as terminology changes.

Review Questions

1. What is a controlled vocabulary?
2. What makes a thesaurus different from an ordinary dictionary?
3. Create one BT, two NTs, and two RTs for the concept Digital Libraries.
4. List the eleven steps involved in constructing a thesaurus.

Chapter 5: Thesaurus Construction: Structure and Maintenance

By the end of this chapter, you should be able to:

- Explain the purpose of a scope note.
- Use USE and UF references correctly.
- Distinguish between alphabetical and classified thesaurus displays.
- Describe the basic tasks involved in thesaurus maintenance.

5.1 Scope Notes

A scope note is a short explanatory statement attached to a term to clarify exactly how that term is to be interpreted or applied within a particular vocabulary. Scope notes are especially useful when a term is ambiguous in everyday language or when it carries a specialised meaning within the subject field.

5.2 USE and UF References

USE directs a searcher or indexer from a non-preferred term toward the preferred term that should be used instead. UF, which stands for Used For, works in the opposite direction: it is displayed underneath the preferred term to show which non-preferred terms lead to it.

EXAMPLE

AUTOMOBILES — UF: Cars. / CARS — USE: Automobiles.

5.3 Alphabetical Display

In an alphabetical display, terms are simply arranged from A to Z. This arrangement is convenient for a user who already knows the specific term they are interested in and wants to look up its relationships quickly.

5.4 Classified Display

In a classified display, terms are grouped together within a broader conceptual structure rather than alphabetically. This arrangement supports browsing, allowing a user to explore related areas of a subject even when they do not yet know the exact term they need.

5.5 Maintaining a Thesaurus

A thesaurus is never a finished product. Because terminology changes over time, thesauri require continuing revision. Typical maintenance tasks include adding new terms as they emerge, revising definitions and scope notes, merging or splitting concepts as understanding of a field develops, updating cross-references, and keeping a documented record of why each decision was made.

5.6 The Role of Software

Thesaurus-management software can support many of the mechanical aspects of this work, including data entry, checking that relationships are consistent, version control, and publication of the finished thesaurus. However, such software supports the intellectual work of the indexer; it does not replace the human judgement required to decide what a term means and how it relates to other terms.

Review Questions

1. Write a scope note for a term that could easily be misunderstood.
2. Create your own USE and UF pair.
3. Give one advantage of an alphabetical display and one advantage of a classified display.

MODULE 3 — PRE-COORDINATE INDEXING SYSTEMS

Chapter 6: Classic Pre-Coordinate Techniques

By the end of this chapter, you should be able to:

- Define pre-coordinate indexing.
- Explain chain indexing.
- Explain cyclic (cycled) indexing.
- Discuss the advantages and limitations of pre-coordinate systems.

6.1 What Is Pre-Coordinate Indexing?

In pre-coordinate indexing, the concepts that make up a subject are combined into a single indexing expression at the moment the document is indexed — that is, before any user ever performs a search. The indexer, not the searcher, is responsible for deciding how the different concepts relate to one another within that expression.

EXAMPLE

University Libraries — Automation — Nigeria. The relationship between these three concepts, and the order in which they appear, is determined entirely by the indexer at the time of indexing.

6.2 Key Characteristics

Pre-coordinate systems can produce carefully controlled and well-structured representations of a document's subject, and are common in traditional subject-heading systems. Their main limitation is that the indexer must try to anticipate which combinations of concepts will actually be useful to future users, which is not always possible to predict with certainty, and skilled, trained indexers are needed to apply the rules correctly.

6.3 Chain Indexing

Chain indexing is closely associated with the Indian librarian and classification theorist S. R. Ranganathan. This technique derives subject entries from the hierarchical chain of concepts found within a classification scheme, and it creates access points directly from that conceptual structure, illustrating the close relationship between classification and subject indexing.

6.4 Cyclic (Cycled) Indexing

Cyclic indexing approaches generate multiple index entries by varying the position or focus of terms within a compound subject, so that each significant component is, in turn, brought into the leading position. This provides several different access points that all lead back to the same document, increasing the chance that a user will find it regardless of which term they search first.

WORKED EXAMPLE

The subject "Use of Artificial Intelligence in University Libraries in Nigeria" could generate cycled access points such as: Artificial Intelligence — University Libraries — Nigeria; University Libraries — Artificial Intelligence — Nigeria; and Nigeria — University Libraries — Artificial Intelligence.

6.5 Advantages and Limitations

Pre-coordinate systems offer several advantages, alongside some real limitations:

- Advantage — controlled and predictable representation of subjects.
- Advantage — explicit relationships between concepts are preserved.
- Advantage — access points are consistent and predictable for users.
- Advantage — can provide highly precise subject access.
- Limitation — considerable intellectual effort is required from the indexer.
- Limitation — the system depends heavily on the rules being applied correctly.
- Limitation — the combinations a user actually searches for may differ from those the indexer anticipated.
- Limitation — users may need to understand the indexing structure to search it effectively.

Review Questions

1. Define pre-coordinate indexing in your own words.
2. Why might pre-coordinate indexing be considered less flexible than post-coordinate indexing?
3. Which scholar is closely associated with chain indexing?
4. What is the purpose of cycling in indexing? Illustrate with an example of your own.

Chapter 7: Permuted and Algorithmic Systems

By the end of this chapter, you should be able to:

- Explain what SLIC is.
- Explain the purpose and origin of PRECIS.
- Understand the role of operators and string generation.

7.1 SLIC (Selective Listing in Combination)

SLIC stands for Selective Listing in Combination. It is a pre-coordinate approach that provides multiple access points by generating selected combinations of the terms assigned to a document, emphasising the analysis of subject concepts and their relationships when constructing index entries.

7.2 PRECIS (Preserved Context Index System)

PRECIS stands for Preserved Context Index System. It was developed by the British indexer Derek Austin for the British National Bibliography, and was designed specifically to preserve the context and the relationships between concepts within a subject statement while still generating a set of usable index entries automatically, once the necessary input had been supplied. Its major contribution was its attempt to preserve the meaning and context of concepts, rather than merely listing isolated keywords, and it was significant in the wider development of computer-assisted subject indexing.

7.3 Role Operators in PRECIS

Role operators indicate the function that each concept plays within a subject statement — for example, distinguishing an action from its agent, its object, or another role it might play. Because the exact symbols and rules used can differ between manuals, students should always learn and apply the version prescribed by their own lecturer.

7.4 How a PRECIS String Is Built

Building a PRECIS-style index string generally follows a practical sequence:

1. Understand the subject of the document as a whole.
2. Identify the individual concepts contained within that subject.
3. Identify the role that each concept plays.
4. Apply the prescribed role operators to each concept.
5. Generate the finished index string from the ordered, labelled concepts.

7.5 Why Preserving Context Matters

Consider a simple, unordered list of terms: students; libraries; databases; use. This list does not make clear who is using what, or what the actual relationship between these concepts is. A syntactic system such as PRECIS is designed precisely to prevent this kind of ambiguity by preserving the relationships between the concepts.

Review Questions

1. What does SLIC stand for?
2. What does PRECIS stand for, and who developed it?
3. Why are role operators necessary in a system like PRECIS?
4. Why is preserving context useful to a searcher?

MODULE 4 — POST-COORDINATE AND DERIVED INDEXING

Chapter 8: Post-Coordinate Indexing Systems

By the end of this chapter, you should be able to:

- Define post-coordinate indexing.
- Explain how terms are combined at the point of retrieval.
- Describe free-text searching, Uniterm indexing, and optical coincidence systems.

8.1 What Is Post-Coordinate Indexing?

In post-coordinate indexing, concepts are assigned to a document separately, as individual terms, and it is the searcher or the retrieval system that combines them together at the moment of searching, rather than the indexer combining them in advance.

EXAMPLE

A document might simply receive the three separate terms Artificial Intelligence, University Libraries, and Reference Services, without any single combined expression being created in advance. A user could later combine any two or three of these with AND to retrieve it.

8.2 Pre-Coordinate versus Post-Coordinate

The essential difference is timing. Pre-coordinate indexing combines concepts while the document is being indexed. Post-coordinate indexing leaves each term separately searchable and lets the combination happen later, during retrieval.

8.3 Free-Text Searching

Free-text searching looks for words directly within fields such as titles, abstracts, or the full text of a document, without relying entirely on a controlled vocabulary. This approach is flexible, but it can suffer from problems such as synonyms not being matched, ambiguous words, and variations in spelling.

8.4 Uniterm Indexing

Uniterm indexing is an early and influential example of the post-coordinate approach. It works by assigning individual, single terms to documents and then combining those terms only at the point of retrieval, illustrating the basic principle behind all post-coordinate systems.

8.5 Optical Coincidence Systems

Historically, optical coincidence systems used physical cards or other machine-readable formats to identify which documents shared a chosen set of terms. Modern digital Boolean searching performs the same essential logical operations, but does so electronically rather than mechanically.

8.6 Advantages and Limitations

Post-coordinate systems allow for flexible and often unexpected combinations of terms at search time, are well suited to computerised retrieval, and scale well to large databases. However, this flexibility comes at a cost: it places greater responsibility on the searcher to construct an effective query, results can be affected by inconsistent or uncontrolled terminology, and a poorly formed search can produce high recall but low precision.

Review Questions

1. Define post-coordinate indexing in your own words.
2. Give two differences between pre-coordinate and post-coordinate indexing.
3. What is Uniterm indexing?
4. Why can post-coordinate searching produce high recall but low precision if handled poorly?

Chapter 9: Search Strategies and Derived Indexing

By the end of this chapter, you should be able to:

- Explain how a search strategy is formulated.
- Use basic Boolean operators.
- Explain what derived indexing is, including its advantages and disadvantages.
- Describe KWIC and KWOC.

9.1 What Is a Search Strategy?

A search strategy is a planned approach to translating a real information need into searchable concepts, specific terms, and useful combinations of those terms. Searching is generally an iterative activity, meaning that a searcher typically refines their approach over several attempts rather than getting the perfect result in a single step.

9.2 Boolean Operators

Three basic Boolean operators are used to combine search terms:

- AND narrows a search by requiring that both concepts be present.
- OR broadens a search by accepting either concept.
- NOT excludes a specified concept from the results.

EXAMPLE

LIBRARIES AND ARTIFICIAL INTELLIGENCE narrows a search to documents containing both concepts.
CARS OR AUTOMOBILES broadens a search to accept either term.

9.3 Formulating a Search

A typical search formulation follows these steps:

1. Start by identifying the major concepts in the information need.
2. Identify synonyms and related terms for each concept.
3. Run an initial search using these terms.
4. Inspect the records retrieved for useful results and new vocabulary.
5. Refine the query, using controlled vocabulary terms where they are available.

9.4 Derived Indexing

Derived indexing obtains its index terms directly from features of the document itself — most commonly the title, but also abstracts, full text, author-supplied keywords, and captions. This approach can be fast and economical, since it requires little additional intellectual effort and is well suited to automation, but its quality depends entirely on how well the source field actually represents the document's subject.

Because derived terms are taken from the author's own words rather than a controlled vocabulary, they may include irrelevant words, fail to capture implied context, or run into synonym and ambiguity problems that a trained indexer would normally resolve.

9.5 KWIC (Keyword in Context)

KWIC stands for Keyword in Context. In a KWIC display, a significant keyword is shown together with the words surrounding it in the title, so that the keyword retains the context in which it originally appeared.

9.6 KWOC (Keyword Out of Context)

KWOC stands for Keyword Out of Context. In a KWOC display, the keyword is shown separately from the rest of the title, typically as a heading, followed underneath by the full citation.

9.7 Comparing KWIC and KWOC

KWIC keeps each keyword embedded within its surrounding title, preserving context at a glance. KWOC instead places the keyword outside the title as a separate heading. Both techniques create multiple access points from terms that have been derived directly from the document, rather than assigned by an indexer.

Review Questions

1. Build a Boolean search statement for finding documents about social media and academic performance among university students.
2. Explain the difference between KWIC and KWOC.
3. State one advantage and one limitation of derived indexing.
4. Why is derived indexing well suited to automation?

MODULE 5 — PERFORMANCE EVALUATION AND SYSTEM METRICS

Chapter 10: Evaluation Methodologies

By the end of this chapter, you should be able to:

- Explain why indexing systems need to be evaluated.
- Define relevance, exhaustivity, and specificity.
- List the recognised criteria for evaluating an index.
- Explain the significance of the Cranfield research tradition.

10.1 Why Evaluate an Indexing System?

Having a large collection of documents and assigning many index terms does not automatically mean that a retrieval system is actually useful to its users. Evaluation provides real evidence about whether users are able to find relevant information efficiently, rather than relying on assumptions.

10.2 Relevance

A document is considered relevant when it genuinely satisfies a user's information need. It is important to understand that relevance is not the same thing as simple keyword matching — a document can contain the exact word a user searched for and still fail to actually answer that user's question.

10.3 Exhaustivity

Exhaustivity refers to the extent to which the important concepts within a document are represented by the index terms assigned to it. A more exhaustive indexing approach provides more conceptual access points into the same document.

10.4 Specificity

Specificity refers to how precisely an indexing term represents the actual subject of the document. For a document specifically about university libraries, the term "Libraries" is less specific than the term "University Libraries."

10.5 The Trade-Off Between Exhaustivity and Specificity

Exhaustivity asks how many important concepts were represented, while specificity asks how precisely each individual concept was represented. Indexing that is too broad can increase the retrieval of irrelevant material, even though it may appear to be thorough.

10.6 Criteria for Evaluating an Index

Beyond relevance, exhaustivity, and specificity, a comprehensive evaluation of an index examines several further qualities:

- Accuracy — index terms should correctly represent the document's content.
- Consistency — different indexers should produce reasonably similar results when indexing the same material.
- Usability — users should be able to understand and use the index without special training.
- Currency — the vocabulary should reflect current terminology and changes in the subject field.
- Coverage — the index should adequately cover the documents in the collection as a whole.

10.7 The Cranfield Experiments

The Cranfield research programme, closely associated with the researcher Cyril Cleverdon, helped to establish a systematic, evidence-based approach to evaluating retrieval systems. This tradition introduced the use of test collections, sets of queries, and formal relevance judgements, allowing retrieval results to be measured rather than merely asserted. It encouraged experimental comparison between systems, rather than unsupported claims about which indexing approach was best.

10.8 Over-Retrieval and Under-Retrieval

Over-retrieval occurs when a search returns too many irrelevant documents. Under-retrieval occurs when relevant documents are missed altogether. The quality of the indexing, the strength of vocabulary control, and the search strategy used can all influence which of these problems occurs.

Review Questions

1. Define exhaustivity.
2. Define specificity.
3. List five recognised criteria for evaluating an index, beyond exhaustivity and specificity.
4. What was important about the Cranfield research tradition?
5. Explain the difference between over-retrieval and under-retrieval.

Chapter 11: Precision, Recall, and Search Optimisation

By the end of this chapter, you should be able to:

- Define and calculate precision, recall, and fallout.
- Explain the meaning of TP, FP, FN, and TN.
- Explain the trade-off between precision and recall.
- Apply basic techniques of search optimisation.

11.1 Four Possible Outcomes

Every document in a collection, in relation to a given search, falls into one of four possible categories:

- True Positive (TP): the document is relevant and was retrieved.
- False Positive (FP): the document is not relevant, but was retrieved anyway.
- False Negative (FN): the document is relevant, but was not retrieved.
- True Negative (TN): the document is not relevant and was correctly not retrieved.

11.2 Precision

Precision answers the question: of everything that was retrieved, how much of it was actually relevant?

FORMULA

$$\text{Precision} = \text{Relevant Retrieved} \div \text{Total Retrieved} = TP \div (TP + FP)$$

11.3 Recall

Recall answers a different question: of all the relevant documents that exist in the collection, how many of them were actually retrieved?

FORMULA

$$\text{Recall} = \text{Relevant Retrieved} \div \text{Total Relevant} = TP \div (TP + FN)$$

11.4 A Simple Worked Example

Suppose a collection contains 100 relevant documents in total. A search retrieves 20 documents, of which 15 turn out to be relevant.

CALCULATION

Precision = $15 \div 20 = 75\%$. Recall = $15 \div 100 = 15\%$. This search produced fairly clean results, but it missed a large number of the relevant documents that were actually available.

11.5 Fallout

Fallout measures the proportion of non-relevant documents that were retrieved, out of all the non-relevant documents that exist in the collection.

FORMULA

$$\text{Fallout} = \text{Non-Relevant Retrieved} \div \text{Total Non-Relevant} = FP \div (FP + TN)$$

11.6 A Fuller Worked Example

Suppose a collection contains 1,000 documents in total, of which 200 are relevant to a particular search. That search retrieves 100 documents, of which 80 are relevant.

CALCULATION

Precision = $80 \div 100 = 80\%$. Recall = $80 \div 200 = 40\%$. Non-relevant retrieved = 20; total non-relevant = 800; Fallout = $20 \div 800 = 2.5\%$.

11.7 The Precision–Recall Trade-Off

A broad search strategy tends to increase recall, but it often retrieves more irrelevant results along with it, which reduces precision. A narrow search strategy tends to improve precision, but risks missing relevant documents. The appropriate balance between the two depends entirely on the specific information need of the user.

11.8 Improving Recall

- Use synonyms and related terms.
- Broaden the search appropriately.
- Use the OR operator.
- Search additional fields, not only the title.
- Make use of controlled-vocabulary relationships such as BT and NT.
- Inspect useful results for additional relevant terminology.

11.9 Improving Precision

- Use more specific terms.
 - Combine concepts using the AND operator.
 - Use phrase searching where it is supported.
 - Apply suitable filters.
 - Exclude unwanted concepts carefully, using NOT.
-

Review Questions

1. A database contains 50 relevant documents. A search retrieves 25 documents, of which 20 are relevant. Calculate the precision and recall.
2. A database contains 500 documents, 100 of which are relevant. A search retrieves 50 documents, of which 35 are relevant. Calculate the precision, recall, and fallout.
3. Explain, in your own words, what it means for a search to have high precision but low recall.

MODULE 6 — ABSTRACTING PRINCIPLES AND APPLICATIONS

Chapter 12: Abstracting Techniques and Types

By the end of this chapter, you should be able to:

- Define an abstract.
- Explain content extraction and editing.
- Distinguish between indicative, informative, critical, slant, author's, and structured abstracts.
- Write a simple academic abstract.

12.1 What Is an Abstract?

An abstract is a concise representation of a document's content. Depending on its particular type, an abstract may present the purpose of a study, the method used, the main findings, and the conclusion reached.

12.2 Why Abstracts Matter

Abstracts help users screen large amounts of literature quickly and decide which documents genuinely deserve a full reading. They are particularly important within bibliographic and indexing-and-abstracting databases, where a user may be scanning hundreds of records at once.

12.3 Content Extraction

When preparing an abstract, the writer should ask a consistent set of questions of the source document:

- What is the problem being addressed?
- What is the objective of the work?
- What exactly was studied?
- How was it studied — what method was used?
- What were the main findings?
- What conclusion or recommendation was reached?

12.4 Principles of Good Abstracting

A good abstract should be clear, concise, accurate, objective, self-contained, coherent, grammatically correct, and faithful to the source document. It should never introduce unsupported claims, invented information, or the personal opinions of the person writing the abstract.

12.5 The Indicative Abstract

An indicative abstract tells the reader what a document covers and answers the question “What does this document discuss?”, but it usually does not report detailed findings. It is generally brief.

EXAMPLE

This article discusses the application of artificial intelligence in university libraries, examining major uses, potential benefits, implementation challenges, and implications for future services.

12.6 The Informative Abstract

An informative abstract provides substantive information, most commonly including the purpose, method, findings, and conclusion of a study. It is detailed enough that a reader can often use it to judge whether the original document is worth reading in full.

EXAMPLE

This study examined AI adoption among university librarians in Nigeria. A questionnaire was administered to librarians in selected universities. The findings showed high awareness but limited formal training, along with concerns about accuracy and data privacy. The study recommends regular AI-literacy programmes and institutional guidance.

12.7 The Indicative-Informative Abstract

As the name suggests, an indicative-informative abstract combines features of both types above: it indicates the general scope of the document while also reporting some of its substantive findings, without going into the full detail of a purely informative abstract.

12.8 The Critical Abstract

A critical abstract goes a step further than a purely descriptive summary. In addition to summarising the content, it offers an assessment or evaluation of the document — commenting, for example, on the strengths, weaknesses, methodology, reliability, or significance of the work. Because it involves judgement rather than pure description, a critical abstract is usually prepared by someone with subject expertise.

12.9 The Author's Abstract

An author's abstract is simply an abstract prepared by the author of the original document, rather than by an independent indexer or abstractor. Many journals require authors to submit one alongside their article.

12.10 The Slant Abstract

A slant abstract emphasises the particular aspects of a document that are most relevant to a specific audience. The very same source document may be presented differently depending on whether the intended reader is a librarian, an administrator, or a technical specialist.

12.11 The Structured Abstract

A structured abstract organises its content under clear headings, such as Background, Objective, Method, Results, and Conclusion. This format improves scanning and makes it easy for a reader to locate a specific piece of information quickly.

12.12 Editing an Abstract

Editing involves removing repetition, correcting inaccuracies, simplifying unclear wording, checking that the important findings are actually present, and making sure that no personal judgement or invented information has crept into the text.

12.13 A Complete Worked Example

Source material: a study examined electronic database use by postgraduate students at three Nigerian universities. Questionnaires were administered to 300 students, producing 240 valid responses. Seventy-eight percent of respondents used databases at least monthly, while 45 percent had difficulty developing

effective search strategies. Google Scholar and institutional databases were the most frequently used resources, and the researchers recommended information-literacy training.

RESULTING INFORMATIVE ABSTRACT

This study examined electronic database use among postgraduate students at three Nigerian universities. Questionnaires were administered to 300 students, producing 240 valid responses. The findings showed that 78% used electronic databases at least monthly, while 45% experienced difficulty developing effective search strategies. Google Scholar and institutional databases were frequently used. The study recommends regular information-literacy training to strengthen students' database-searching skills.

Review Questions

1. What is the difference between an indicative abstract and an informative abstract?
2. What is a critical abstract, and how does it differ from an author's abstract?
3. What is a slant abstract?
4. List the common headings used in a structured abstract.
5. List five qualities of a good abstract.

Chapter 13: Uses and Applications of Abstracts

By the end of this chapter, you should be able to:

- Explain the major uses of abstracts.
- Explain Current Awareness Services (CAS) and Selective Dissemination of Information (SDI).
- Describe how abstracts function within bibliographic databases and major indexing-and-abstracting services.
- Explain the difference between extractive and abstractive summarisation.
- Discuss the relationship between human and AI-assisted abstracting.

13.1 Abstracts in Databases

Bibliographic and indexing-and-abstracting databases typically contain titles, authors, publication details, assigned indexing terms, and abstracts. The abstract helps a user judge the relevance of a document before deciding whether to obtain the full text.

13.2 Current Awareness Services (CAS)

Current Awareness Services keep users informed about newly available information within a subject area of interest to them. Alerts sent through such services may include citations together with abstracts, allowing users to quickly identify newly published material that is relevant to their work.

13.3 Selective Dissemination of Information (SDI)

Selective Dissemination of Information distributes newly available material according to the individual interests or profile of each user. Abstracts help recipients of an SDI service decide quickly whether a newly identified document is actually useful to them.

13.4 Abstracts and Information Retrieval

A single search can return hundreds of records. Reading the abstracts attached to those records allows a user to screen them efficiently, rather than having to read every full document, and to select only the most promising ones for closer reading.

13.5 Major Indexing and Abstracting Services

Indexing and abstracting services provide organised access to scholarly and professional literature across disciplines, combining the indexing and abstracting principles covered throughout this guide into large-scale, continuously updated products. Well-known examples include Scopus, Web of Science, PubMed/MEDLINE, the ERIC database, Chemical Abstracts, Library and Information Science Abstracts (LISA), Library Literature & Information Science Full Text, and the EBSCO and ProQuest families of databases. These services assist researchers in discovering literature across different disciplines, and are themselves a practical demonstration of subject indexing and abstracting at scale.

13.6 Automated Summarisation

Automated summarisation uses computational methods to produce shorter representations of documents. It becomes especially useful when very large collections of material must be processed quickly, at a scale that would be impractical for human abstractors alone.

13.7 Extractive Summarisation

Extractive methods work by selecting important words, phrases, or whole sentences directly from the source document. The resulting summary is therefore built largely from text that already existed in the original.

13.8 Abstractive Summarisation

Abstractive methods attempt to represent the meaning of the source document using newly generated wording, rather than simply lifting text from the original. Modern AI systems can assist with this kind of summarisation, but their output may sometimes omit important details or distort the original meaning, so it always requires careful checking against the source.

13.9 Human Abstractors versus Automated Systems

Human abstractors bring contextual judgement to their work and can tailor an abstract for a specialised audience in a way that automated systems currently cannot match. Automated systems, in turn, offer speed and the ability to work at a much larger scale. In practice, a workflow that combines automated assistance with human quality control often produces the best results.

13.10 The Complete Workflow

The relationship between indexing and abstracting can be summarised as a single continuous workflow: a document undergoes subject analysis, is indexed, is abstracted, becomes part of a database record, is found through a user's search, is screened by the user through its record, and is finally selected for full-text reading. Indexing provides the structured access points that make a document findable in the first place; abstracting then provides the concise representation of content that helps the user decide whether it is worth reading in full.

Review Questions

1. What is CAS, and how does it differ from SDI?
2. How do abstracts help the users of a database?
3. Name three major indexing and abstracting services and the disciplines they primarily serve.

4. Explain the difference between extractive and abstractive summarisation.
5. Why should AI-generated abstracts always be checked against the original source?

MODULE 7 — DIGITAL, AUTOMATED AND AI-ASSISTED INDEXING

Chapter 14: Automatic Indexing, Computer-Assisted Tools and Artificial Intelligence

By the end of this chapter, you should be able to:

- Explain automatic and computer-assisted indexing.
- Describe hyperlinked and dynamic indexes in digital publishing.
- Discuss what Artificial Intelligence can and cannot currently do in indexing and abstracting.
- Explain the continuing role of the human indexer and abstractor.

14.1 From Manual to Digital Indexing

Traditionally, indexing was carried out entirely by hand: indexers read whole manuscripts, identified important concepts, prepared index cards or handwritten notes, arranged entries alphabetically, and assigned page references. This process was highly labour-intensive, time-consuming, and susceptible to human error. The digital publishing revolution has introduced computer-assisted indexing tools, electronic publishing platforms, digital books, online databases, and Artificial Intelligence, all of which have significantly changed indexing practice. Although the fundamental principles — subject analysis, concept identification, consistency, specificity, and user-centred organisation — remain unchanged, the methods used to apply them have become more sophisticated and efficient.

14.2 Automatic Indexing

Automatic indexing uses computer algorithms to identify and assign terms to a document by analysing its titles, abstracts, full text, keywords, metadata, citations, and author information, without an indexer manually assigning every term. Its advantages include rapid processing, the ability to handle large-scale indexing that would overwhelm a purely manual process, reduced manual labour, consistency of application, and suitability for very large collections. Its limitations are equally real: algorithmic context may be misunderstood, ambiguous terms may be incorrectly interpreted, meaningful semantic relationships may be missed altogether, and human quality control is still generally necessary before the results are relied upon.

14.3 Computer-Assisted Indexing Software

A distinct category of tool assists, rather than replaces, the professional indexer. Computer-assisted indexing software helps create accurate, consistent, and well-organised indexes while reducing the time and effort required for repetitive tasks, automating routine functions such as sorting entries alphabetically, checking for duplicate entries, maintaining consistent terminology, managing cross-references, and formatting the finished index to a publisher's specifications. Among the most widely recognised packages are CINDEX, a professional indexing tool widely used by publishers and freelance indexers; Sky Index, a Windows-based application that supports large indexing projects while automatically maintaining alphabetical order and page references; and Macrex, one of the earliest professional indexing programs, particularly popular in academic publishing for its flexibility with large scholarly works and multiple levels of subentries. These tools increase productivity by automating formatting and organisational tasks, freeing the indexer to concentrate on the intellectual work of deciding what deserves an entry.

14.4 Hyperlinked and Dynamic Indexes

Electronic books, online textbooks, digital encyclopaedias, institutional repositories, and web-based publications are no longer restricted to static page numbers. Instead, they often use hyperlinks, bookmarks, anchors, and search functions in place of conventional locators. In a hyperlinked index, an entry links directly to the relevant section of the document, so a reader can click through to the required information immediately rather than manually turning to an indicated page. Some digital publications go further still, using dynamic indexes that update automatically whenever the underlying document is revised — unlike a printed index, which must be completely reworked after any change to pagination. Digital indexes may also link to figures, tables, appendices, multimedia resources such as audio and video, external websites, related chapters, and supplementary online materials, so that a modern digital index has evolved from a simple alphabetical list into an interactive navigation system.

14.5 What Artificial Intelligence Can Do

AI-powered systems apply Natural Language Processing (NLP), Machine Learning (ML), semantic analysis, and text mining to analyse documents and identify important concepts. In indexing and abstracting work more broadly, AI can assist with:

- extracting keywords automatically;
- identifying frequently occurring concepts;
- recognising the names of people, organisations, and places (named-entity recognition);
- detecting relationships among concepts and performing semantic indexing;
- recommending indexing terms and generating preliminary index entries;
- identifying synonyms and related expressions;
- classifying topics according to subject categories;
- suggesting appropriate cross-references;
- generating metadata and assisting with automatic abstracting; and
- supporting similarity detection, recommendation, and question-answering systems built on top of indexed collections.

These capabilities substantially reduce the time required for preliminary indexing and abstracting, particularly when dealing with lengthy academic publications or very large digital collections.

14.6 The Continuing Role of the Human Indexer

Despite these advances, Artificial Intelligence has not replaced the professional indexer or abstractor. Indexing remains fundamentally an intellectual activity that requires human judgement, contextual understanding, subject expertise, and sensitivity to users' information needs. AI systems can readily identify frequently occurring words, but they often struggle to determine which concepts are truly significant, meanings that are implied rather than stated, the difference between important and trivial information, the contextual meaning of an ambiguous term, the appropriate level of specificity, likely user search behaviour, and conceptual relationships that call for professional interpretation.

EXAMPLE

An AI system may notice the word “cloud” hundreds of times in a computer science textbook. A professional indexer recognises that the relevant concept is Cloud Computing, not simply “cloud,” and may divide the entry into subentries such as cloud infrastructure, cloud security, and cloud storage. Likewise, a history book may repeatedly mention “Washington” — a human indexer can tell whether this means George Washington, Washington, D.C., or Washington State, ensuring readers are directed accurately.

14.7 Benefits and Challenges of Digital and AI-Assisted Indexing

The adoption of digital technologies has brought real benefits: faster index preparation, improved consistency of entries, automatic alphabetical arrangement, efficient management of cross-references, easier correction and revision, better collaboration between publishers and indexers across locations, support for very large indexing projects, interactive hyperlinks in electronic publications, reduced production costs, and improved accessibility through searchable, navigable indexes.

These gains come with real challenges. AI-generated indexes may include irrelevant or excessive entries, because algorithms often rely on word frequency rather than genuine conceptual importance; automated systems may overlook implicit ideas, fail to recognise specialised terminology, or misinterpret context. Other recurring challenges include over-reliance on automated tools, inconsistencies in AI-generated terminology, software compatibility issues across publishing platforms, the need for continual software updates and staff training, difficulty indexing multimedia and interactive content, and ethical concerns relating to algorithmic bias and transparency. These limitations reinforce the importance of professional human oversight throughout the indexing and abstracting process.

14.8 Future Trends

The future of indexing and abstracting is expected to involve deeper integration of Artificial Intelligence, semantic technologies, and linked data: AI systems that understand context more accurately, semantic indexing that works with concepts rather than isolated keywords, multilingual indexing for global audiences, voice-search-compatible indexes, and personalised digital indexes that adapt to individual users' interests and search histories. Cloud computing and collaborative online platforms are also enabling indexers, editors, and publishers to work simultaneously on the same projects regardless of location. As digital libraries and open-access repositories continue to expand, indexing and abstracting will increasingly support the discovery and retrieval of information across vast collections of interconnected resources — with the fundamental principles set out in this guide continuing to underpin that work, whichever tools are used to apply them.

Review Questions

1. What is automatic indexing, and what are its main advantages and limitations?
2. Name three computer-assisted indexing software packages and explain what they automate.
3. Explain the difference between a hyperlinked index and a dynamic index.
4. List five tasks that AI can assist with in indexing and abstracting.
5. Using the Cloud Computing or Washington example (or one of your own), explain why human judgement remains necessary in indexing.
6. State three challenges of digital or AI-assisted indexing.

Appendix A: Practical Exercises

Thesaurus Construction

1. Construct a thesaurus entry for the term University Libraries, including one Broader Term, two Narrower Terms, two Related Terms, and one UF reference.
2. Construct a thesaurus entry for the term Information Literacy, and justify each relationship you establish.

Indexing Practice

Index the subject “Use of social media by postgraduate students for academic communication.” Identify at least four concepts, and explain the reasoning behind each choice.

Further Indexing Practice

For each title below, propose at least four appropriate index terms and briefly justify your choices:

1. “Use of Social Media by University Students for Academic Research in Nigeria.”
2. “Application of Artificial Intelligence in Academic Libraries.”
3. “Preservation of Indigenous Knowledge Resources in Nigerian University Libraries.”

PRECIS Preparation

Analyse the statement “Postgraduate students use electronic databases for academic research.” Identify the agent, the action, the object, and the purpose within this statement before applying the exact operator rules prescribed by your own course.

KWIC and KWOC Practice

Create both a KWIC entry and a KWOC entry for the title “Artificial Intelligence Applications in Nigerian University Libraries.”

Evaluation Practice

A collection contains 800 documents, of which 160 are relevant to a particular search. That search retrieves 100 documents, of which 70 are relevant. Calculate the precision, recall, and fallout for this search.

Abstracting Practice

Write a 60–80 word indicative abstract and a separate 100–150 word informative abstract for a hypothetical paper about information-literacy training and database use among postgraduate students.

Full Abstracting Exercise

Working from a research article of approximately 1,500–3,000 words (your own choice, or one supplied by your lecturer):

1. Identify the title.
2. Identify the research problem.
3. Identify the objectives.
4. Identify the methodology.
5. Identify the major findings.
6. Identify the conclusion.

7. Prepare a 150–250 word informative abstract.
8. Identify five appropriate keywords.

Appendix B: Revision Questions and Model Answers

Short-Answer Questions

1. Define subject indexing.
2. What is a controlled vocabulary?
3. Define BT, NT, and RT.
4. What is pre-coordinate indexing?
5. What is post-coordinate indexing?
6. What do KWIC and KWOC stand for?
7. Define exhaustivity and specificity.
8. State the precision and recall formulas.
9. Define indicative and informative abstracts.
10. What is CAS?
11. What is an index, and how does it differ from a table of contents?
12. Who developed PRECIS, and what does the acronym stand for?
13. What is automatic indexing?
14. Name two computer-assisted indexing software packages.
15. What is a critical abstract?

Model Answers

- Subject indexing represents the subject content of a document for the purpose of retrieval.
- A controlled vocabulary is an organised set of authorised terms used consistently for indexing and retrieval.
- BT is Broader Term, NT is Narrower Term, and RT is Related Term.
- Pre-coordinate indexing combines concepts before retrieval takes place.
- Post-coordinate indexing combines separate concepts during the act of retrieval.
- KWIC is Keyword in Context; KWOC is Keyword Out of Context.
- Exhaustivity concerns the number of important concepts represented; specificity concerns how precisely each one is represented.
- Precision equals relevant retrieved divided by total retrieved. Recall equals relevant retrieved divided by total relevant.
- Indicative abstracts describe the scope of a document; informative abstracts provide substantive information and findings.
- CAS is a Current Awareness Service.
- An index is an alphabetical guide to specific information within a document, usually placed at the end; a table of contents appears at the start and lists chapters in the order they occur.
- PRECIS, the Preserved Context Index System, was developed by Derek Austin.
- Automatic indexing uses computer algorithms to identify and assign index terms to a document with little or no manual intervention.
- CINDEX, Sky Index, and Macrex are widely used computer-assisted indexing software packages.
- A critical abstract summarises a document's content while also offering an assessment of its strengths, weaknesses, or significance.

Essay Questions

1. Discuss the objectives and guiding principles of subject indexing.
2. Compare pre-coordinate and post-coordinate indexing.
3. Explain thesaurus structure and semantic relationships.
4. Discuss the contribution of the Cranfield tradition.
5. Explain precision, recall, and fallout, using examples.
6. Compare indicative and informative abstracts, and explain where the critical and slant abstract fit alongside them.
7. Discuss the role of abstracts in Current Awareness Services, Selective Dissemination of Information, and database searching.
8. Discuss the advantages and limitations of AI-assisted indexing and abstracting.
9. Discuss the process and challenges of book indexing.
10. Examine the role of computer-assisted indexing software and Artificial Intelligence in modern indexing services.

A Model Comparison Outline

When answering a comparison question, such as one contrasting pre-coordinate and post-coordinate indexing, the following outline provides a reliable structure:

1. Introduction: define both systems clearly.
2. Timing: pre-coordinate combines concepts during indexing; post-coordinate combines them during retrieval.
3. Roles: pre-coordinate places more responsibility on the indexer; post-coordinate gives the searcher more responsibility for combining concepts.
4. Flexibility: post-coordinate generally permits more novel combinations of terms.
5. Control: pre-coordinate can provide carefully designed, controlled conceptual expressions.
6. Limitations: pre-coordinate indexing may fail to anticipate every combination a user might want; post-coordinate indexing depends heavily on the searcher's own skill and use of terminology.
7. Conclusion: the most suitable system depends on the particular information environment and retrieval objectives involved.

Appendix C: Glossary of Key Terms

Abstract — A concise representation of a document's content.

Abstracting — The process of preparing abstracts.

Assigned indexing — Indexing in which an indexer selects terms from an indexing language or controlled vocabulary, rather than copying them directly from the document.

Author's abstract — An abstract prepared by the author of the original document.

Automatic indexing — Indexing performed by computer algorithms with little or no manual intervention.

BT — Broader Term; a more general concept within a thesaurus hierarchy.

Book index — An alphabetical guide to the significant content of a book, with locators, usually placed at the end.

CAS — Current Awareness Service.

Chain indexing — A pre-coordinate indexing technique, associated with S. R. Ranganathan, that derives subject entries from the hierarchical chain of a classification scheme.

Computer-assisted indexing — The use of software to support tasks such as sorting, formatting, and cross-reference management, while the indexer retains intellectual control.

Controlled vocabulary — An organised set of authorised terms used consistently for indexing and retrieval.

Critical abstract — An abstract that summarises a document while also offering an evaluation of its strengths, weaknesses, or significance.

Cross-reference — A See or See also note directing a reader from one index term to another.

Cyclic (cycled) indexing — A technique that varies the leading term in a compound subject to create multiple access points.

Derived indexing — Indexing based on features taken directly from the document, such as title words.

Dynamic index — A digital index that updates automatically when the underlying document changes.

Exhaustivity — The extent to which important document concepts are represented in indexing.

Fallout — The proportion of non-relevant documents retrieved, out of all non-relevant documents available.

Hyperlinked index — A digital index whose entries link directly to the relevant section of a document.

Index — A systematically arranged guide to the contents of a document, consisting of terms and locators.

Indexing — Representing document content with terms or symbols to support retrieval.

Indexing and abstracting service — A database or publication, such as Scopus or MEDLINE, that provides organised indexing and abstracting access to literature across a discipline.

Indicative abstract — An abstract that indicates the scope or topics covered by a document.

Informative abstract — An abstract that reports substantive content such as purpose, method, findings, and conclusion.

ISO 999 / BS ISO 5963 — International and British standards providing guidance on the content, structure, and preparation of indexes.

KWIC — Keyword in Context.

KWOC — Keyword Out of Context.

Locator — The page, section, or paragraph reference attached to an index entry.

NT — Narrower Term; a more specific concept within a thesaurus hierarchy.

Post-coordinate indexing — Indexing in which concepts are combined during retrieval.

PRECIS — Preserved Context Index System, developed by Derek Austin.

Precision — The proportion of retrieved documents that are relevant.

Pre-coordinate indexing — Indexing in which concepts are combined before retrieval.

Recall — The proportion of all relevant documents that are actually retrieved.

RT — Related Term; an associated concept that is neither equivalent nor hierarchical.

SDI — Selective Dissemination of Information.

SLIC — Selective Listing in Combination.

Specificity — The precision with which an indexing term represents a concept.

Subject analysis — The intellectual examination of a document to identify its subject content.

Thesaurus — A controlled vocabulary organised by defined semantic relationships.

UF — Used For; a reference from a preferred term to a non-preferred term.

USE — A reference directing a non-preferred term to the preferred term.

Assessment and Examination Guide

Assessment	Weight	Main Preparation
Indexing Practicum	20%	Thesaurus entry design, book-index construction, and PRECIS exercises
Abstracting Portfolio	15%	Indicative, informative, and critical abstracts
Mid-Term Assessment	15%	Pre-coordinate versus post-coordinate analysis
Final Examination	50%	Comprehensive theory and practical examination, including digital and AI-assisted indexing

Final Revision Checklist

- Explain the major terms of this course in your own words.
- Cite at least one scholarly definition of indexing and of an index (e.g. Lancaster, Mulvany, Knight, or the relevant ISO standard).
- Construct basic thesaurus relationships (BT, NT, RT, USE, UF).
- Distinguish clearly between pre-coordinate, post-coordinate, and derived indexing.
- Explain PRECIS, its origin, its syntax, and its role operators.
- Construct simple KWIC and KWOC displays.
- Describe the components and types of a book index, and the stages of book indexing.
- Calculate precision, recall, and fallout.
- Explain exhaustivity, specificity, and the other criteria used to evaluate an index.
- Write indicative, informative, and critical abstracts.
- Compare human and automated approaches to indexing and abstracting, including the role of AI.
- Answer practical questions in your own words, without relying on memorised wording.